

Mini Project 2 - Gallatin Housing Predictions

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Executive Summary

Gallatin County, especially the Bozeman area, has seen a sharp rise in home prices over the past few decades. This report examines the main factors influencing single-family home prices using sales data from more than 17,000 properties across the county. The goal is to understand what drives price differences and how that information can guide development, investment, and housing policy.

What We Found

Home location, size, and features are the strongest predictors of price. Homes in Big Sky stand out as the most expensive, largely due to location and access to Lone Peak High School, one of the county's highest-rated schools. Across Gallatin County, homes with more bathrooms, larger living areas, and bigger lots consistently sold for higher prices, showing that buyers place strong value on space, amenities, and school quality.

What It Means for Affordability and Investment

The high demand for housing in areas like Big Sky and Bozeman is making it harder for local residents to afford new homes. Nearby communities, such as Belgrade, Manhattan, and Three Forks are found to offer more affordable options. However, these communities are in areas with less desirable schools and fewer housing amenities. Therefore, the nearby communities could attract more home buyers and investors if school and housing amenities were improved and more available.

Recommendations

- **Developers:** Focus new housing development in areas where prices are still affordable for home buyers. Offer smaller lot size or townhome designs to meet the growing local need.
- **Policymakers:** Support balanced growth with development zoning flexibility and investments in schools.
- **Investors:** To build long-term value, invest in properties near highly rated schools and in growing areas like Big Sky and Bozeman. These investments are likely to hold long-term value.

In summary, we found that location and home features are significant factors in the Gallatin County housing market. By planning growth and development with these factors in mind, local leaders can help improve housing affordability while maintaining a healthy real estate market.

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1. Introduction

Gallatin County, Montana, home to the rapidly growing city of Bozeman, has faced significant challenges over the past several years, primarily driven by surging housing demand and limited supply. This confluence of factors, exacerbated by the greater Bozeman area's desirability as a destination for recreation, education, and quality of life, has led to a dramatic and sustained increase in single-family home prices, creating an affordability crisis for many residents. As individuals living in this community, we are keenly aware of the impact these housing market dynamics have on local infrastructure, economy, and social fabric.

Our analysis focuses on predicting single-family home prices in Gallatin County based on several key property and locational attributes. These include details about the home's structure, such as the number of bedrooms and bathrooms, the total living space, and the lot size. Furthermore, we consider the age of the structure, its geographic location within the county, and the closeness and quality of local high schools.

The following sections detail our methods, including data cleaning, feature extraction, Exploratory Data Analysis (EDA), and model development, to explore the relationship of housing prices in Gallatin County with county (i.e., school and city) and home features.

2. Methods & Results

2.1 Data Cleaning and Feature Extraction

A dataset containing 25,378 single-family home sales in Gallatin County, Montana, was obtained from Zillow. We cleaned the dataset by selecting variables relevant to predicting home prices (Table 1). We noticed that several cities do not have consistent naming conventions in the data set (e.g., West Yellowstone was also named W Yellowstone or west yellowstone). We applied a REGEX correction to reformat the names so that unique cities were consistently named.

To identify potential outliers due to erroneous entries, we plotted box-and-whisker plots for each numeric variable and scatter plots of each variable against price to visually inspect points far from the data's mean and central cluster. The box-and-whisker plots were too noisy, so we used observations from our scatter plots to omit data lying outside the central cluster ranges (Fig. 1).

Next, we examined the dataset for missing values and identified only three observations with missing entries. Because of this small number, we chose to omit these observations. Categorical variables (e.g., city) were transformed using one-hot encoding, as they represent nominal data without inherent order. Numeric variables (e.g., lot size, bedrooms, living area) were standardized using z-score scaling to ensure consistent scaling across features.

We also extracted additional features to enrich the dataset. These included the nearest high school and its rating for each home. The high school names were one-hot encoded, and the school ratings were standardized using the same z-score method as other numeric features. We noted that Gallatin High School lacked ratings, resulting in NaN values for all its entries. Because these ratings could not be imputed or transformed, we excluded the corresponding entries. We also checked for outliers in the high school and school rating data by applying the upper and lower quartiles to trim the dataset further. The final cleaned and processed dataset contained 17,954 single-family home sales, representing approximately 70% of the original data.

Table 1. Table of data variables. Included (highlighted in green) and excluded (highlighted in orange) variables in the Gallatin County single-family home dataset.

Variable	Reason for Inclusion	Reason for Exclusion
unnamed		Not relevant to home prices.
zpid		Not relevant to home prices.
state		The dataset only contains MT and not other states.
homeStatus		Not relevant to home prices.
streetAddress		Not relevant, this is unique to each home.
homeType		Not relevant, the dataset only contains single-family home types.
dateSoldString		Not relevant to home prices.
photoCount		Not relevant to home prices.
parcelID		Not relevant to home prices.
daysOnZillow		Not relevant to home prices.
dateSold		Not relevant to home prices.
city	Some cities are more desirable places to live.	
bedrooms/bathrooms	Generally increases home value.	
price	This is our response variable.	
yearBuilt	It can influence home value.	
zipcode	Some zip codes are more desirable to live in.	
Lat/Long	Some areas are more desirable to live in.	
livingArea	Home size can influence home value.	
schools	People with children tend to want to live near better schools	
interior/interior_full	Extra home information that can influence price.	

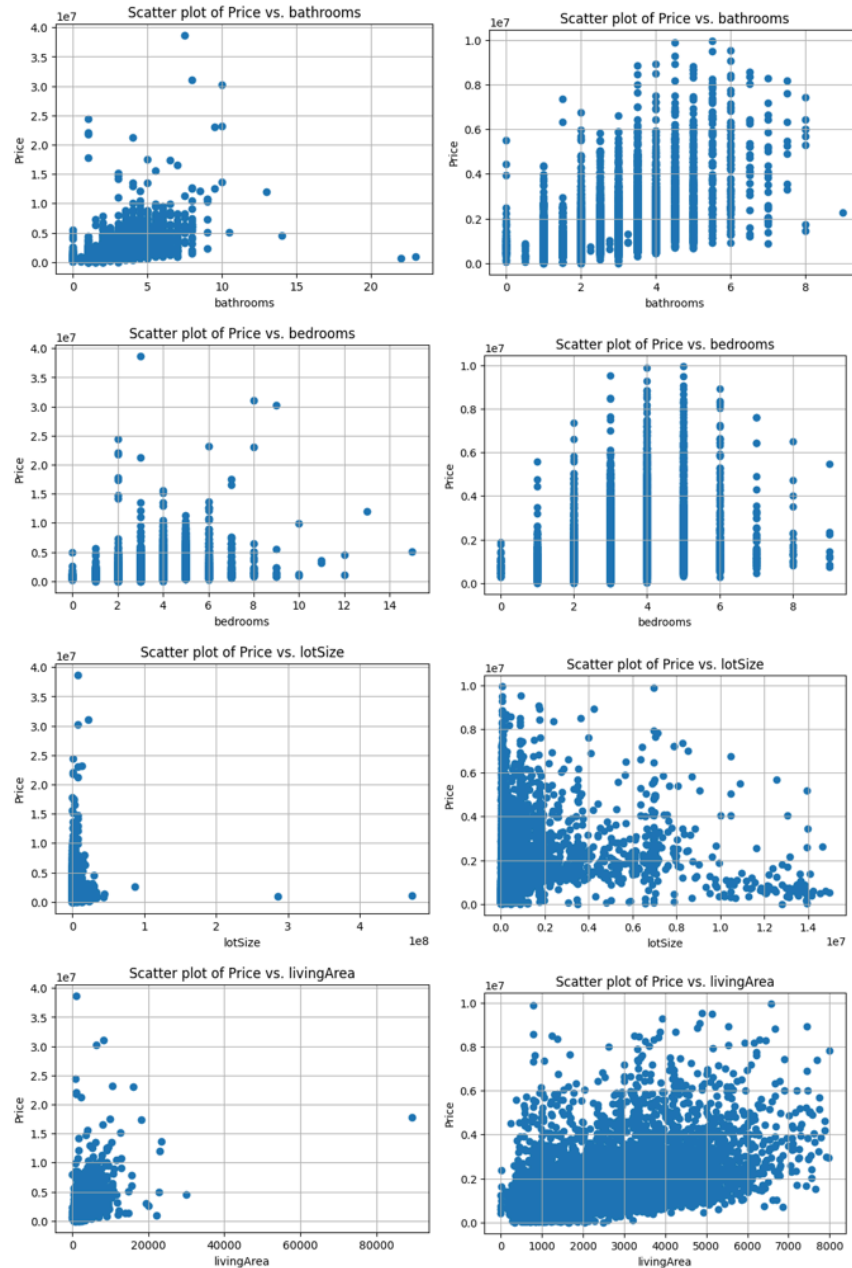


Fig. 1. Outlier detection scatter plots. Scatter plots in the left column show data before outlier removal. Scatter plots in the right column show data after outlier removal.

2.2 Exploratory Data Analysis

To understand the relationships among variables and home prices, and how they may interact or influence one another, we conducted exploratory data analysis (EDA). We began by formulating the following questions to uncover insights into the dataset:

1. Do highly rated high schools explain single-family home prices?
2. Do the number of bedrooms and bathrooms influence single-family home prices?

3. Does the city influence home prices?

To address these questions, we examined variable correlations and collinearity. We also conducted a Principal Component Analysis (PCA) to identify which variables explained the greatest variance in the dataset. To assess the spatial clustering of home prices across Gallatin County, we plotted home prices on a map. Finally, we analyzed price relationships by plotting each key variable against the home price to visually inspect how these variables changed as the price increased.

From the analysis, we noted several interesting findings to our questions.

1. High school ratings were moderately correlated with home prices ($r = +0.41$), indicating that higher school ratings are associated with higher home prices. The school with the strongest relationship was Lone Peak High School ($r = +0.49$), located in Big Sky, MT (Fig. 2). We do make note here that the location might be a confounding variable for the correlation.
2. The number of bathrooms had a stronger influence on home prices than the number of bedrooms, as supported by PCA loadings, correlations ($r = +0.58$ compared to $r = +0.25$, respectively), and box-and-whisker plots (Figs. 2 & 6).
3. City location also played an important role, with Big Sky showing the highest correlation with home prices ($r = +0.51$) and clustering of high home prices compared to other cities in Gallatin County (correlations $< +0.12$; Fig. 2).

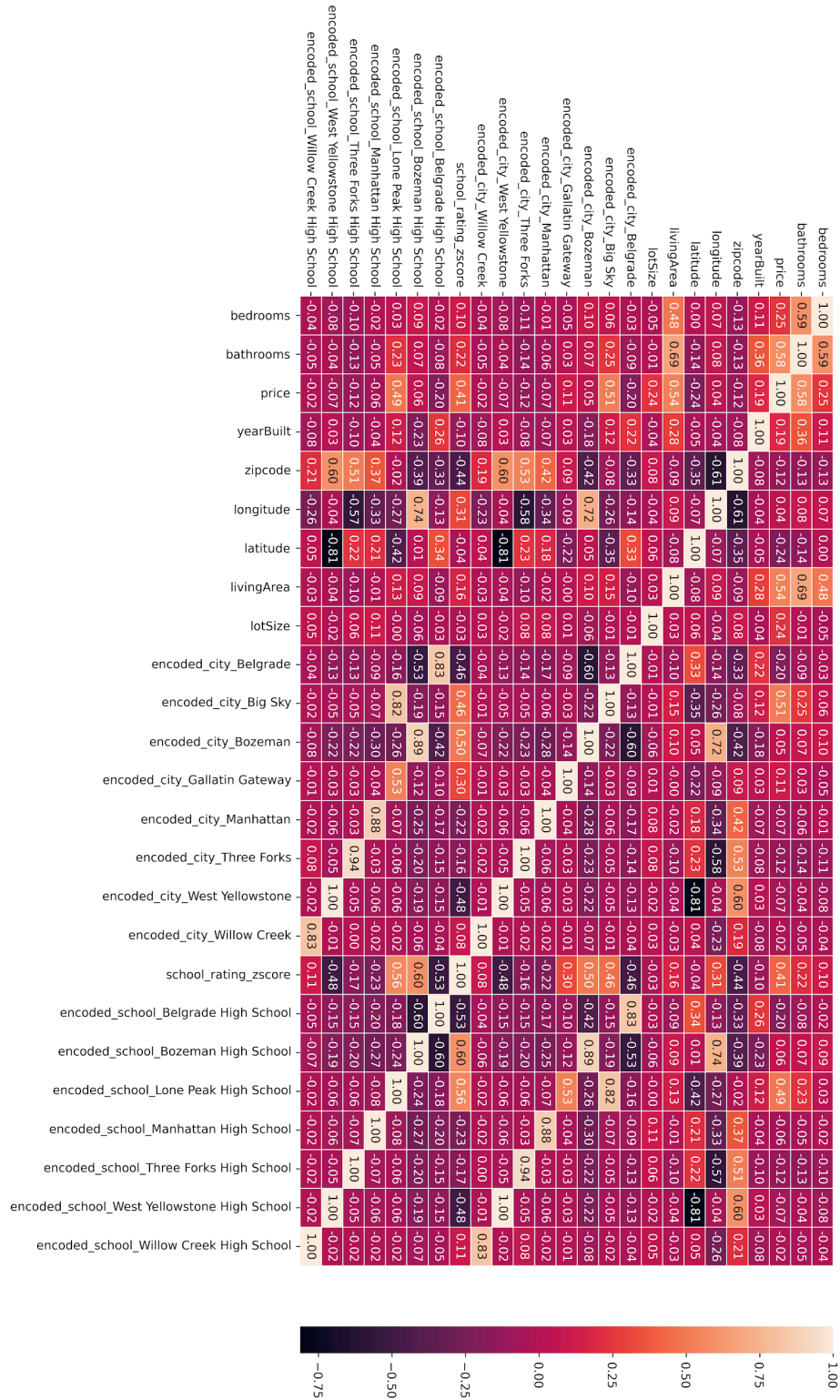


Fig. 2. Correlation heatmap of single-family home variables. Color ranges from dark red (negative correlation) to light peach (high correlation).

Single Family Home Prices in Gallatin County, MT

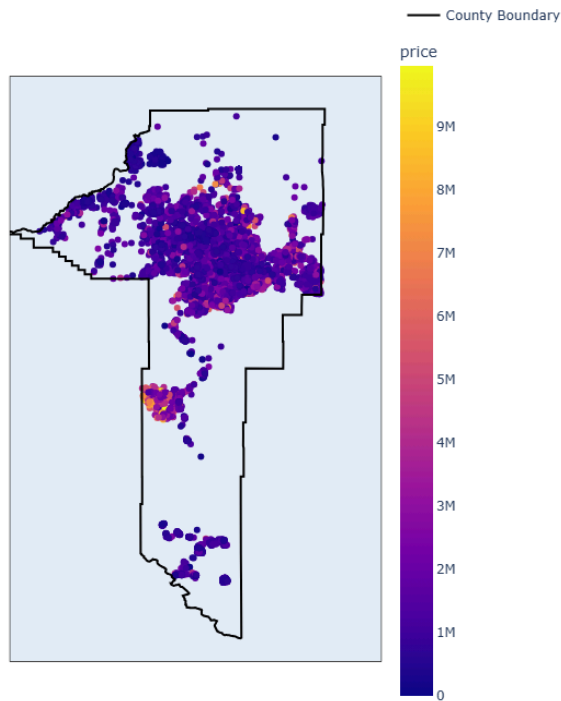


Fig. 4. Spatial map of Gallatin County single-family home prices. Color gradient from dark purple (low home price) to yellow (high home price).

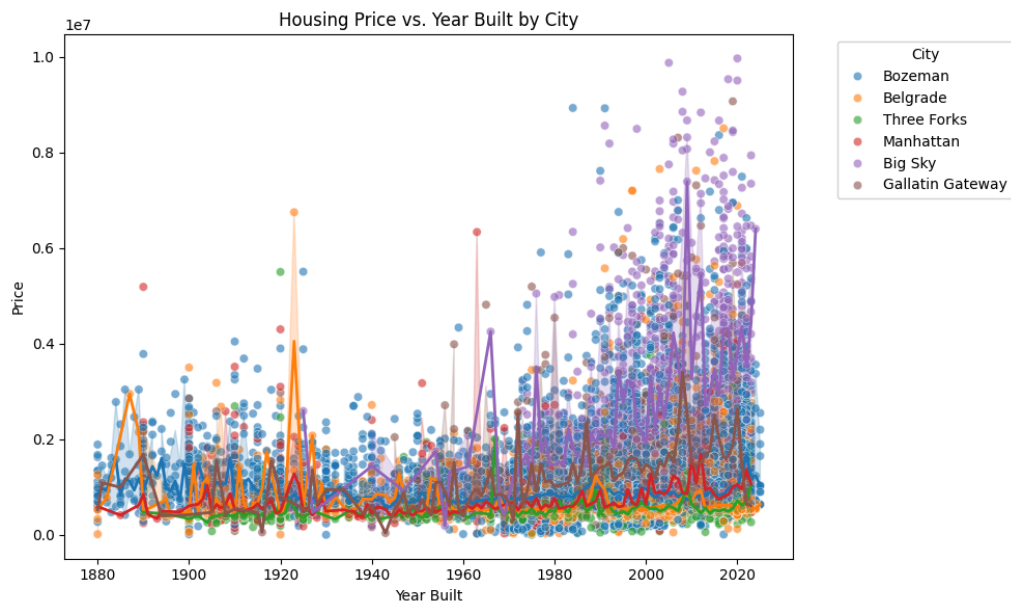


Fig. 5. Line plot of home prices by year built in each city. Unique colors represent individual cities in Gallatin County. As we approach 1980, prices in Big Sky and Bozeman began to climb, while in Belgrade, Manhattan, and Three Forks remained relatively stable.

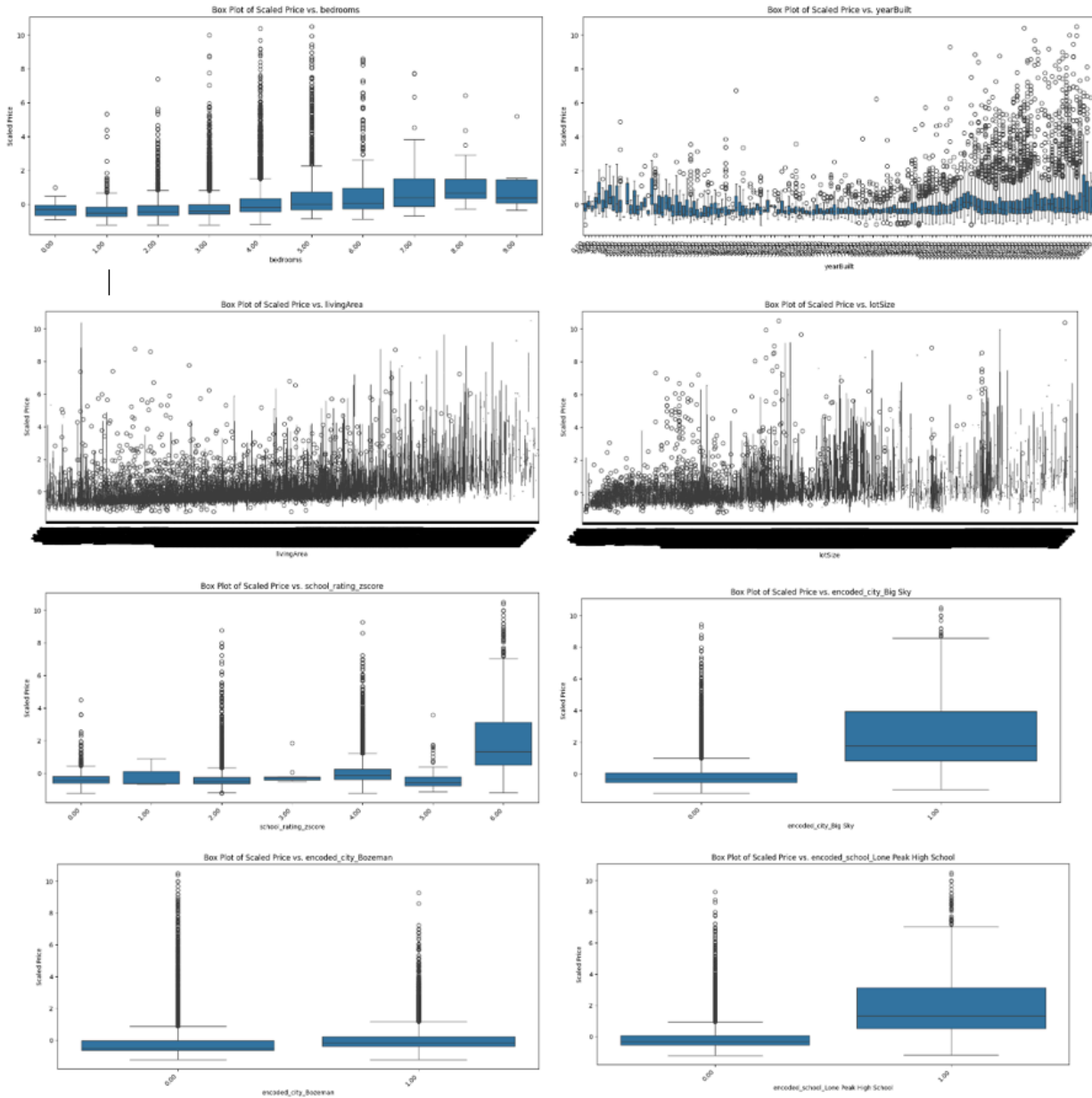


Fig. 6. Box-and-whisker and line plots of select variables shown to increase or decrease as home prices change.

Other variables explaining variation in the dataset driving home prices were identified in the collinearity analysis and PCA, map, and box-and-whisker plots (Table 2, Figs. 2-6). Specifically, the variables of zipcode, longitude, bathrooms, school rating, living area, city, bedroom, year built, and lot size. Based on observations from our EDA, we developed several predictive models to quantify the relationships between home prices and their most influential features.

Table 2. Collinearity of variables. Collinearity values < 5 (highlighted in blue) indicate there is minimal collinearity and should be considered as influencing variables. We reduced multicollinearity in Lasso and Ridge Regression by using penalty terms.

Variable	Collinearity
High School	Inf
City	Inf
School rating	2835.28
zipcode	177.12
latitude	36.05
longitude	9.54
bathrooms	2.64
Living area	1.97
bedrooms	1.67
Year built	1.39
Lot size	1.06

2.3 Model Development

Before model development, the data was prepared for training and testing. The initial model included the predictor variables of bathrooms, bedrooms, year built, living area, and lot size. The dataset was split into 80% for training and 20% for testing.

We employed several supervised machine learning models to test and train our data. During model training, we used 5-fold cross-validation (CV) for any method that required tuning parameters (CV setup provided with its respective code below). Models that did not use CV for training included Single and Multiple Linear Regression (Sections 2.3.1 & 2.3.2). Models using CV for training included forward subset selection (Section 2.3.1), regularization methods (Lasso and Ridge regression; Sections 2.3.4.1 & 2.3.4.2), Principal Component Regression (PCR; Section 2.3.5), and non-linear models (Section 2.3.6). Cross-validation helped evaluate model performance and identify the model configuration with the lowest Root Mean Squared Error (RMSE) for final selection.

All regression techniques (Sections 2.3.1-2.3.6) were implemented using the scikit-learn (sklearn) library. For regression summary statistics, we also used the statsmodels API, which provides F-, t-, and p-values for assessing the statistical significance of coefficients. This dual approach allowed for both predictive accuracy assessment (via sklearn) and interpretability of model parameters (via statsmodels).

2.3.1 Simple Linear Regression

For Simple Linear Regression, we trained and tested each of the five variables individually to assess their predictive power for home prices. We noted that features of bathrooms and living area produced the smallest RMSE value (< 1.0) and higher coefficients (> 0.5), while bedrooms,

year built, and lot size had higher RMSE values (> 1.0) and lower coefficients (< 0.5 ; Table 3). This indicates that, for the simple linear regression, individual variables of bathrooms and living area had the greatest predictive power for home prices.

Table 3. Single Linear Regression intercepts, coefficients, and RMSE values for each variable. Cells highlighted in blue indicate variables with highest coefficients and lowest RMSE values.

Variable	Intercept	Coefficient	RMSE
Bedrooms	0.0114	0.2571	1.0925
Bathrooms	0.0363	0.5875	0.9140
Year built	0.0364	0.1833	1.1042
Living area	0.0314	0.5624	0.9547
Lot size	0.0021	0.2496	1.1011

2.3.2 Multiple Linear Regression

Multiple Linear Regression added complexity and flexibility to the model by including all variables' contributions to predicting home prices. We noticed a change in coefficient values: some are now negative, while others remain positive and relatively strong. From the coefficient values, bathrooms appear to have the strongest influence on home prices, followed by living area and lot size, while bedrooms and year built have a weak influence and are overtaken by other variables (Table 4). The RMSE value also decreased (0.8434), indicating better accuracy and predictive power than Simple Linear Regression.

Table 4. Multiple Linear Regression intercept and coefficient values for all variables taken into consideration. Cells highlighted in blue indicate variables with higher coefficients.

Variable	Coefficient
Intercept	0.0312
Bedrooms	-0.1649
Bathrooms	0.5018
Year built	-0.0413
Living area	0.2963
Lot size	0.2344

2.3.3 Forward Subset Selection

We used a Forward Subset Selection method with 5-fold CV to determine a subset of three optimal features. The subset consisted of bathrooms, living area, and lot size (Table 5). During this process, we noted that the RMSE decreased again to an average of 0.8000, compared to Linear and Multiple Regression.

Table 5. Forward Subset Selection with 5-fold cross-validation RMSE values. Blue highlighted cell indicates the optimal three features selected by the forward subset selection.

Subset	RMSE
Bathrooms	0.8637
Bathrooms, lot size	0.8240
Bathrooms, living area, lot size	0.8000
Bathrooms, lot size, living area, bedrooms	0.7892
Bathrooms, lot size, living area, bedrooms, year built	0.7882

2.3.4 Regularization

We applied Lasso and Ridge Regularization methods with 5-fold CV to our model. This allowed us to further optimize performance and accuracy by applying the tuning parameter (α) to the three features identified as optimal via forward subset selection. After tuning the model with five different values for the tuning parameter for both methods, we determined the optimal α to be 0.001 (Table 6; Fig. 7).

Table 6. Lasso and Ridge Regression RMSE values by tuning parameter. Blue highlighted cell indicates the optimal α tuning parameter between the two methods.

Tuning Parameter (α)	Lasso RMSE	Ridge RMSE
0.001	0.7872	0.7872
0.01	0.7877	0.7872
0.1	0.8113	0.7872
1	1.0535	0.7872
10	1.0535	0.7872

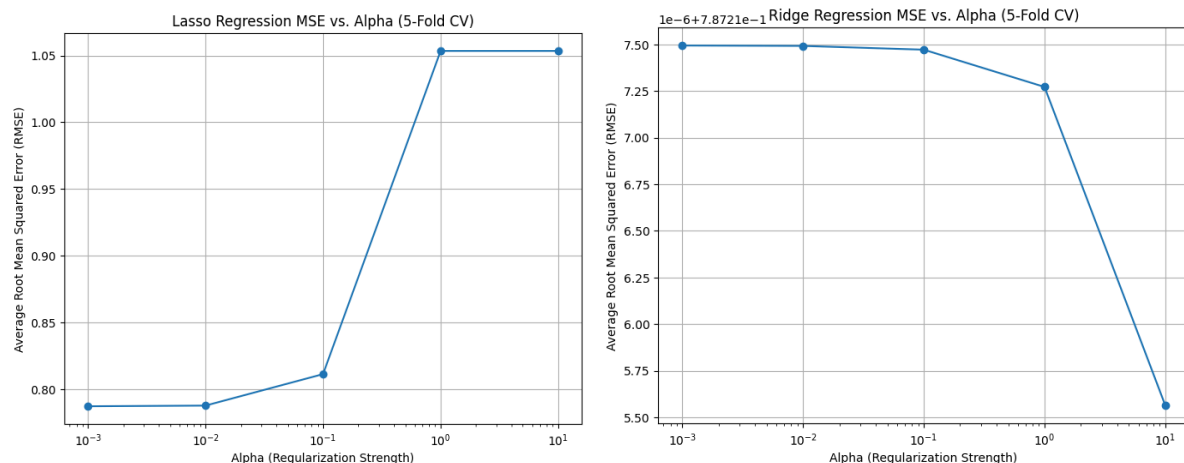


Fig. 7. Lasso and Ridge average RMSE value after 5-fold cross-validation.

2.3.5 Principal Component Regression

We explored, trained, and tested our data using Principal Component Regression using a 5-fold CV to manage multicollinearity and dimensionality. The purpose of this method is to determine the optimal number of components (variables) that explain more than 90% of the variance in our data, with an acceptable RMSE relative to other models (Table 7).

Table 7. Principal Component Regression using a 5-fold CV feature selection RMSE and variance values based on the number of components. Blue highlighted cell indicates the number of components determined acceptable with a low RMSE and high explained variance.

Number of Components	RMSE	Explained Variance (%)
1	0.8938	46.14
2	0.8492	66.12
3	0.8452	85.00
4	0.8047	94.54
5	0.7888	99.99

We selected the first four principal components, which together explained 94.5% of the variance, and used these as predictors in a linear regression model. **This produced a final RMSE of 0.8666**, which is higher than the RMSEs from the previous linear models.

2.3.6 Non-Linear Models

Finally, we explored model development using a polynomial regression using 5-fold CV. We applied this model to all five of the original features at the start of model development. We explored this application using 2nd-, 3rd-, and 4th-degree polynomials. All polynomials across all variables yielded high RMSE values, except for the bathroom variable, which had RMSE values comparable to those of the linear models across all degrees (Table 8). This indicates that the Gallatin Housing dataset's predictive power is mostly linear by nature, with bathrooms showing a partial non-linear relationship.

Table 8. Polynomial Regression RMSE results by variable and degree. Blue highlighted cells indicate possible non-linearity of the variable to home prices.

Variable	Polynomial Degree	Average RMSE
Bedrooms	2	1.0339
Bedrooms	3	1.0327
Bedrooms	4	1.0329
Bathrooms	2	0.8316
Bathrooms	3	0.8250
Bathrooms	4	0.8196
Year built	2	1.0399
Year built	3	1.0399
Year built	4	1.0400

Living area	2	0.8866
Living area	3	0.8789
Living area	4	0.8786
Lot size	2	0.9891
Lot size	3	0.9751
Lot size	4	0.9717

2.3.7 Model Performance Comparison

To determine the final model, we compared Root Mean Squared Error (RMSE) values across all models (Fig. 8). The Lasso and Ridge Regression models achieved the lowest RMSE (~0.7872), indicating the best predictive accuracy for this dataset. These regularized models introduce a penalty term to reduce overfitting, improving generalization while maintaining interpretability. In particular, Lasso Regression enhances interpretability by shrinking some coefficients to zero, effectively performing feature selection.

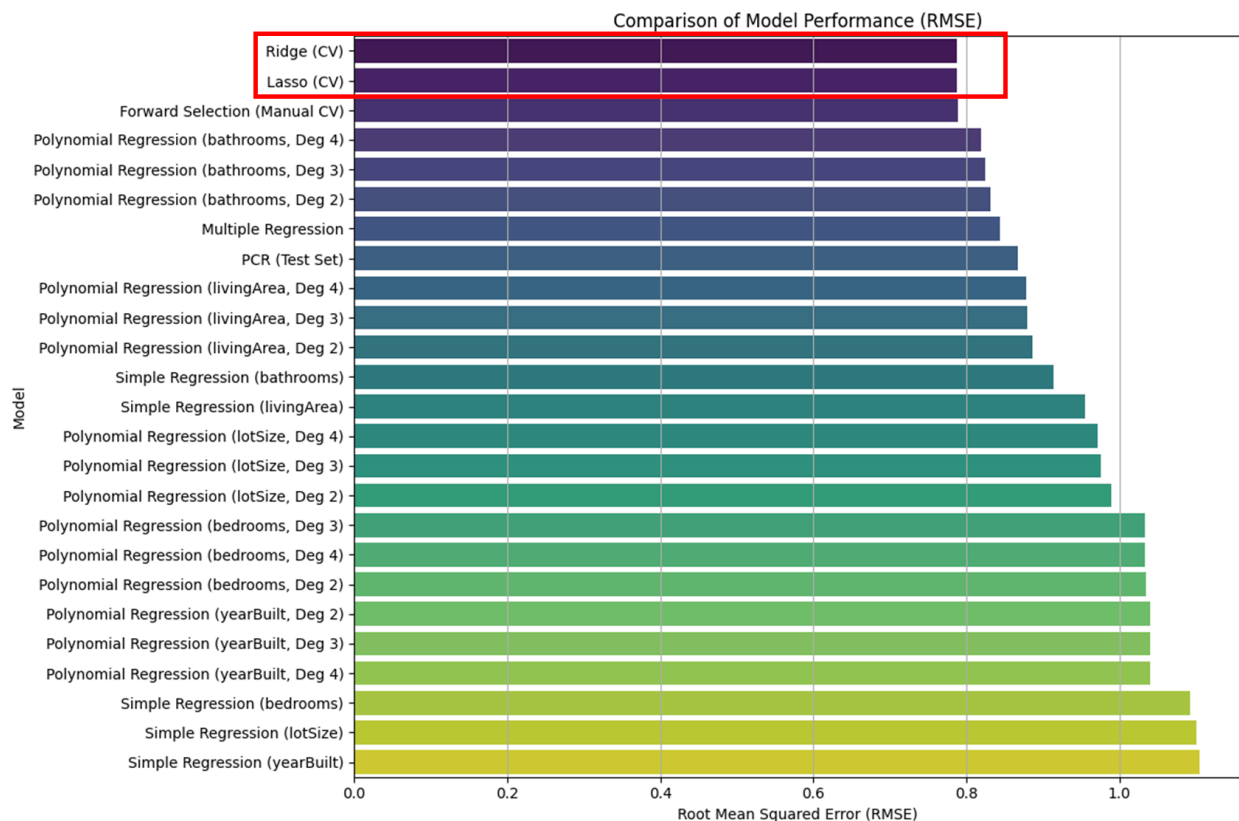


Fig. 8. RMSE comparison between all models developed. The red box indicates our final model selection.

The forward subset selection method performed slightly worse than Ridge and Lasso but better than the unregularized linear models. It offers a strong trade-off between interpretability and

accuracy, producing a standard linear regression with a reduced set of influential features, making it easy to interpret which variables drive home prices.

The multiple linear regression model performed moderately, with a higher RMSE than the regularized or subset models. While more complex than simple regression, it remains interpretable since its coefficients directly quantify each feature's contribution to price under linear assumptions and limited multicollinearity.

The Principal Component Regression (PCR) model produced the highest RMSE among the linear models (excluding simple regression). Although it mitigates multicollinearity by transforming correlated predictors into orthogonal principal components, interpretability is reduced because these components are combinations of multiple features rather than direct variables.

Polynomial regression on individual predictors increased model complexity substantially but did not consistently improve performance. While it can capture nonlinear trends, interpreting higher-order terms is less intuitive, and overfitting can occur without sufficient data regularization.

Our overall comparison shows that increasing model complexity from simple to multiple linear regression improved accuracy but added complexity (though polynomial terms or dimensionality reduction did not yield additional gains). Ridge and Lasso Regression provided the optimal balance between predictive accuracy, model simplicity, and interpretability. Among them, Lasso is particularly valuable for identifying key predictors, as it selects the optimal subset of features most relevant to price variation.

2.4 Model Enhancement and Advanced Predictive Methods

2.4.1 Self-Organizing Map

To explore non-linear relationships and patterns not captured by traditional regression, we applied a Self-Organizing Map (SOM). This unsupervised neural network projects high-dimensional data onto a low-dimensional grid for visualization and clustering. The SOM was trained on the scaled numerical features of the dataset (excluding *price*) to identify groups of similar properties and visualize how features collectively relate to home prices.

The U-matrix (Fig. 9) illustrates distances between neurons: darker areas indicate clusters of similar homes, while lighter areas mark boundaries between clusters. When property prices were overlaid, higher-priced homes (yellow) and lower-priced homes (blue) formed distinct regions, suggesting that the SOM effectively captured price-related structure.

By comparing component planes for individual features (Fig. 10) with the U-matrix, we observed that regions of high prices aligned with high values of *bathrooms*, *bedrooms*, *yearBuilt*,

livingArea, *school_rating*, *city_Big Sky*, and *Lone Peak High School*. This indicates that both property characteristics and location features drive clustering patterns.

Overall, the SOM provided qualitative insight into how combinations of features influence price variation across Gallatin County, highlighting feature groupings that may inform further feature engineering or model refinement.

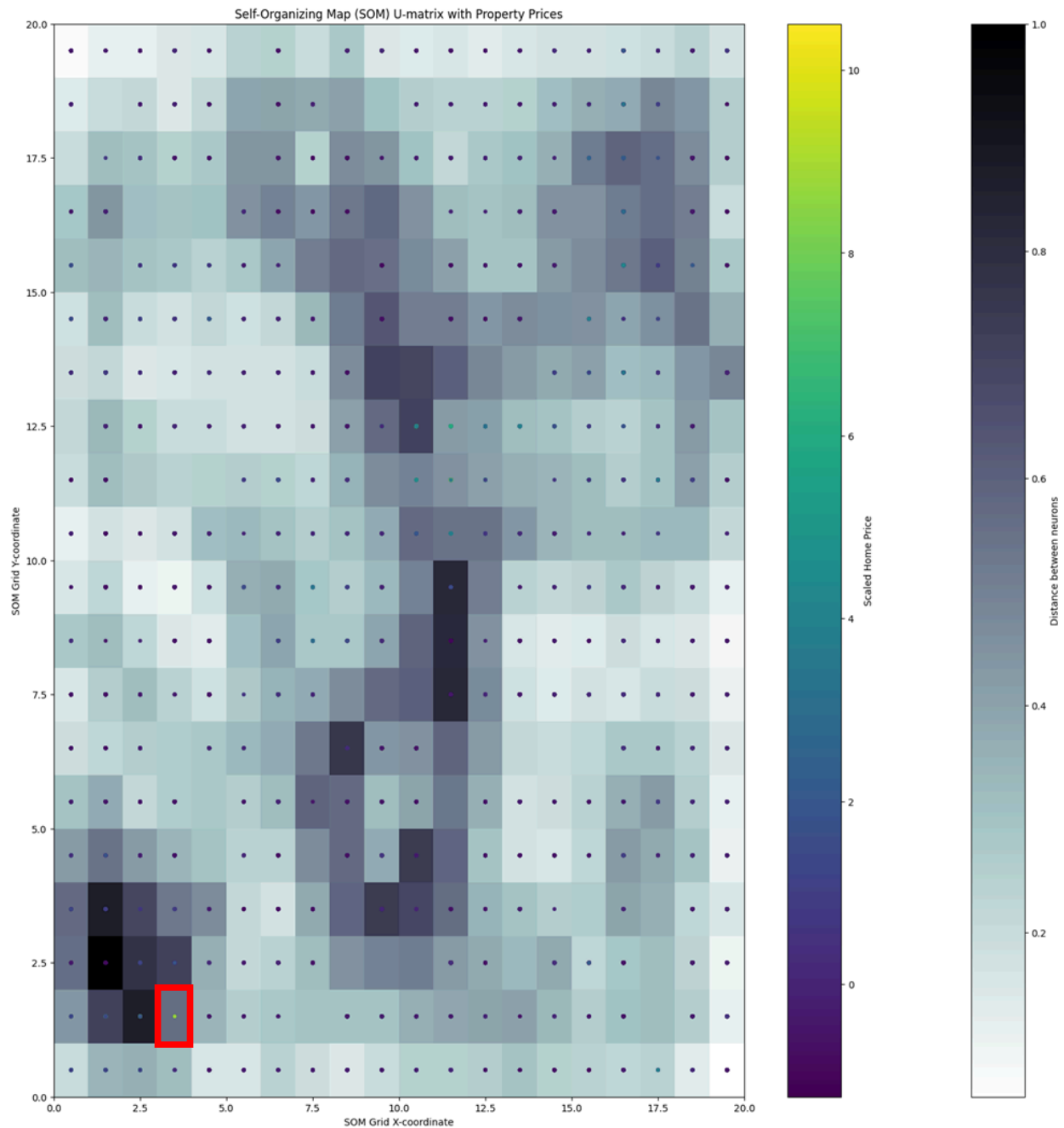


Fig. 9. Self-organizing map U-Matrix grid. The grid shows the distance between neurons with light colors (low distance) to dark colors (high distance). Housing prices are marked as points in

each grid with dark purple (low house prices) to yellow (high house prices). The grid with the red box (coordinates 3, 2) shows a cluster of moderate neuron distance and higher home prices.

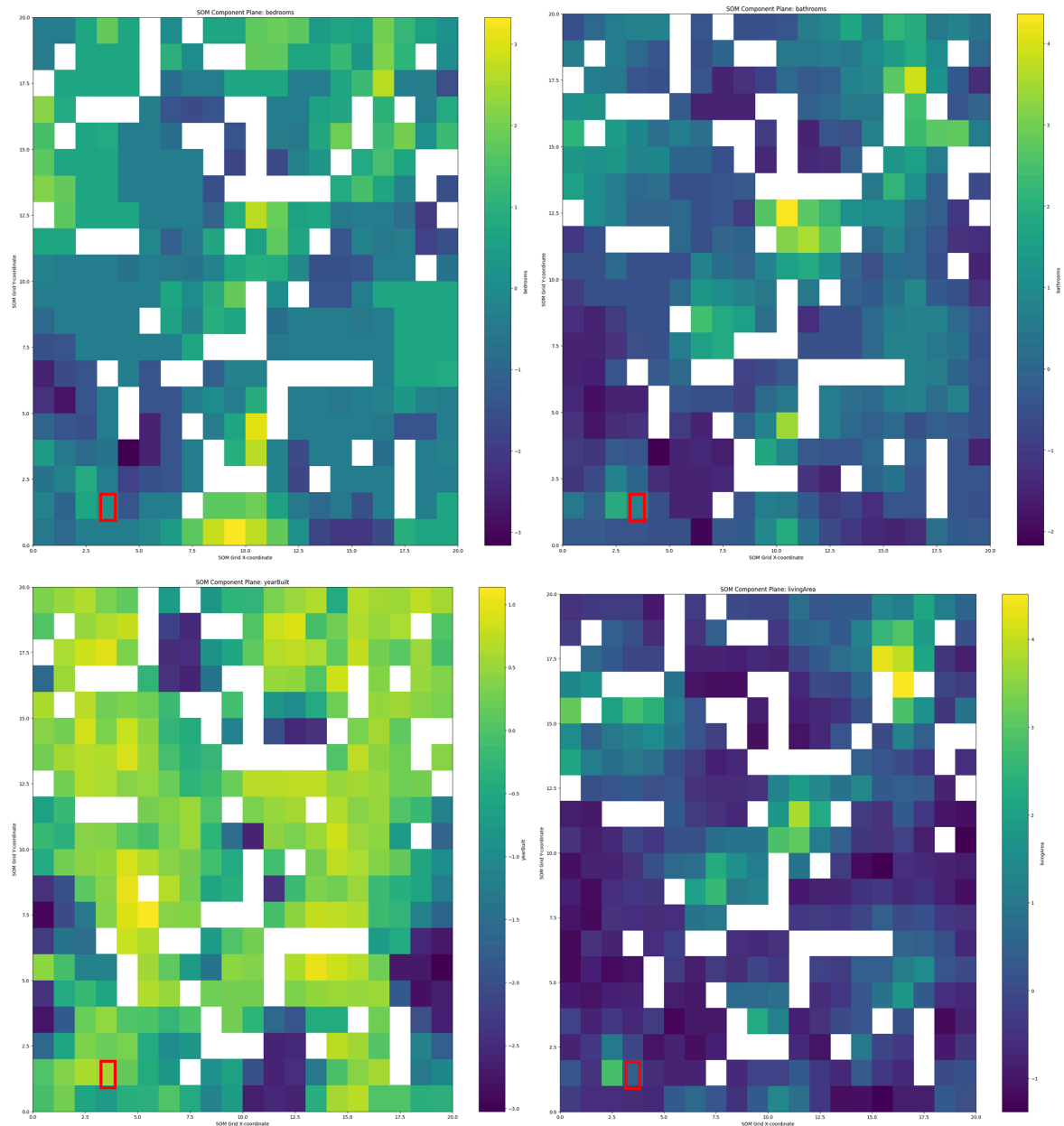


Fig. 10. Component planes for bathrooms, bedrooms, year built, and living area. Other component planes can be viewed in the adjoining Jupyter Notebook. Grid locations of interest are surrounded with a red box for easier comparison to the U-Matrix.

2.4.2 Random Forest Classification

We implemented a Random Forest (RF) model to classify the dataset and assess feature importance, continuing our exploration of the Gallatin Housing data. Random Forest is an ensemble learning method that constructs multiple decision trees using different subsets of the data and predictors, then aggregates their outputs to produce a more stable and accurate prediction. This method is particularly well-suited for non-linear and high-dimensional data, where interactions between predictors may be complex or not well captured by linear models.

The RF model was trained on the same preprocessed dataset used in the SOM. Each tree was built from a random subset of observations and features (bagging), reducing variance and minimizing the risk of overfitting. Model parameters, such as the number of estimators and the maximum tree depth, were tuned via cross-validation to balance bias and variance. The model also provides feature importance scores, which quantify each variable's contribution to reducing prediction error across all trees.

Bathrooms, lotSize, and livingArea emerged as the most important predictors of home price category (Fig. 11), indicating that the property's physical characteristics play a dominant role in classification accuracy. Location-based variables such as encoded_city_Big Sky, longitude, and school_rating_zscore also contributed meaningfully, suggesting that both structural and locational factors drive housing price differentiation in Gallatin County.

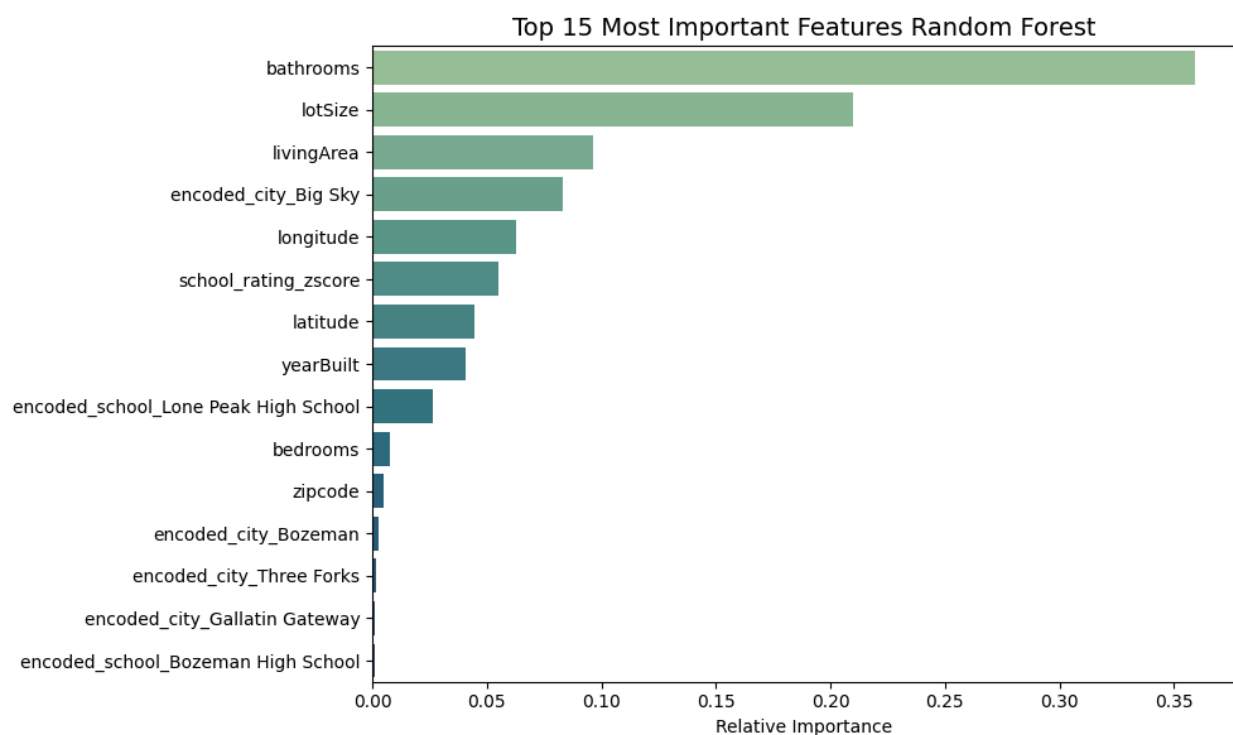


Fig. 11. Random Forest classification of the 15 most important features.

Overall, the Random Forest model corroborated the regression analyses by highlighting the strong predictive power of variables on home prices while providing a more flexible, non-parametric framework for exploring complex relationships among features.

2.5 Final Model Selection & Results

After comparing model performances in Section 2.3.7, we selected Lasso Regularization as the final model for predicting home prices. We included the additional features encoded_city_Big Sky, school_rating_zscore, and encoded_school_Lone Peak High School, which were consistently identified as influential by the PCA, SOM, and Random Forest analyses. Conversely, we excluded zipcode, latitude, and longitude because the city of Big Sky, captured by multiple methods, already captures locational effects correlated with these variables.

The model was tuned using 5-fold cross-validation, resulting in an optimal penalty parameter of $\alpha = 0.001$. At this small value, the Lasso model behaves similarly to an Ordinary Least Squares (OLS) regression, with minimal shrinkage applied to the coefficients. To interpret model significance and feature influence, we therefore used an OLS summary table to approximate the Lasso model's statistical properties.

Feature selection improved model accuracy and interpretability by emphasizing the most influential predictors of home prices in Gallatin County. The final Lasso model indicated that all selected features were statistically significant (p -values ≤ 0.001). Positive coefficients and high t -statistic values for bathrooms, livingArea, lotSize, encoded_city_Big Sky, school_rating_zscore, and encoded_school_Lone Peak High School had the strongest effects on price, with encoded_city_Big Sky showing the greatest influence ($\beta = 1.3161$), followed by encoded_school_Lone Peak High School ($\beta = 0.4794$; Table 9).

In summary, the final Lasso model demonstrated strong predictive accuracy and provided clear insight into the key factors driving home prices in Gallatin County. By isolating the most influential variables, particularly location (Big Sky and its associated school), school quality, and property size metrics, the model offers an interpretable and data-driven understanding of how local housing values are formed. These findings not only validate the importance of locational and structural characteristics but also establish a solid foundation for further analysis of housing affordability and market dynamics across the county.

Table 9. Ordinary Least Squares summary table of intercept, coefficients, and statistical results.

OLS Regression Results						
=====						
Dep. Variable:	y	R-squared:	0.608			
Model:	OLS	Adj. R-squared:	0.608			
Method:	Least Squares	F-statistic:	2781.			
Date:	Thu, 06 Nov 2025	Prob (F-statistic):	0.00			
Time:	17:07:58	Log-Likelihood:	-14685.			
No. Observations:	14363	AIC:	2.939e+04			
Df Residuals:	14354	BIC:	2.946e+04			
Df Model:	8					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	-0.0562	0.006	-9.329	0.000	-0.068	-0.044
x1	-0.1067	0.007	-15.561	0.000	-0.120	-0.093
x2	0.3345	0.009	37.805	0.000	0.317	0.352
x3	-0.0196	0.006	-3.312	0.001	-0.031	-0.008
x4	0.3085	0.008	39.639	0.000	0.293	0.324
x5	0.2321	0.005	44.099	0.000	0.222	0.242
x6	1.3161	0.049	27.071	0.000	1.221	1.411
x7	0.1445	0.007	20.388	0.000	0.131	0.158
x8	0.4794	0.044	10.860	0.000	0.393	0.566
=====						
Omnibus:	8761.674	Durbin-Watson:	1.999			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	227132.231			
Skew:	2.476	Prob(JB):	0.00			
Kurtosis:	21.842	Cond. No.	17.5			
=====						

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

3. Conclusion

Based on the analysis performed, several features consistently emerged as highly important and influential drivers of single-family home prices in Gallatin County (including Bozeman), as detailed across the **Exploratory Data Analysis (Section 2.2)** and **Final Model Selection & Results (Section 2.5)** sections.

3.1 Key Features Driving Home Prices

The final Lasso Regularization model, which incorporated the most influential features identified across all analyses, showed that the following variables had the strongest positive impact on home price:

3.1.1. Location and School Quality

Location-based features proved to have the most substantial impact on home prices, particularly those related to the highly desirable Big Sky area:

- **City Location (encoded_city_Big Sky):** The presence of the Big Sky indicator in the final model showed the greatest influence on price, with a coefficient of $\beta = 1.3161$ (Table 9). This aligns with the EDA, where Big Sky showed the highest correlation with home prices ($r = +0.51$) and distinct spatial clustering of high prices (Fig. 4).
- **School Association (encoded_school_Lone Peak High School):** Closely related to the Big Sky location, the presence of Lone Peak High School was the second strongest predictor, with a coefficient of $\beta = 0.4794$ (Table 9). This corroborates the EDA finding that Lone Peak High School had the strongest individual relationship with home prices ($r = +0.49$; Fig. 2).
- **General School Rating (school_rating_zscore):** The standardized school rating was moderately correlated with home prices ($r = +0.41$; Fig. 2) and was a statistically significant predictor in the final model (Table 9), emphasizing that the quality of local schools is a major factor for buyers.

3.1.2. Property Characteristics

Physical attributes of the home also exerted a significant positive influence on price:

- **Bathrooms:** The number of bathrooms consistently appeared as a primary driver. In Simple Linear Regression, it had the highest coefficient ($\beta = 0.5875$) and lowest RMSE (Table 3). Additionally, it was identified as a key feature in the Forward Subset Selection (Table 5), PCA (Fig. 3), and Random Forest Classification (Fig. 11).
- **Living Area:** The home's size (living area) was a major factor in all analyses, exhibiting a strong positive coefficient in Simple Linear Regression ($\beta = 0.5624$; Table 3) and featuring prominently in the optimal feature subsets selected by both Forward Subset Selection and Random Forest (Table 5, Fig. 11).
- **Lot Size:** The lot size also had a significant positive impact, as evidenced by its inclusion in the optimal three-feature subset (Table 5) and its high importance score in the Random Forest model (Fig. 11).

In summary, the most influential features driving Gallatin County home prices are those related to **desirable location (Big Sky and its associated school)** and **key property size metrics (bathrooms, living area, and lot size)**.

3.2 Limitations and Future Work

While our analysis successfully identified key drivers of single-family home prices in Gallatin County, the study has several limitations that suggest avenues for future research and model enhancement.

3.2.1. Data Quality and Omitted Variables

- **Reliance on Zillow Data:** The dataset's primary source, Zillow, may introduce inherent biases or inaccuracies in reported *price* and *yearBuilt*. Zillow estimates (Zestimates) are not actual appraisals and can influence market perception, while reported property features can be incomplete or outdated.
- **Missing Time-Series Context:** Our model is a cross-sectional analysis of sales, lacking the critical time-series component of sales date (*dateSold*). This omission means the model cannot capture the rapid, significant price appreciation and changing market dynamics that have characterized the Gallatin housing market over the analysis period.
 - **Future Work:** A significant improvement would be to incorporate the sale date as a feature, allowing for the modeling of time-dependent price changes and market-wide appreciation. This would require time-series models (e.g., ARIMA or Prophet) or adding a time-dependent feature to the current regression framework.
- **Omitted Economic and Amenity Variables:** The analysis lacked crucial external economic indicators (e.g., local employment rates, mortgage interest rates, specific property tax rates) and hyper-local amenity data. Variables such as proximity to downtown Bozeman, trail systems, public lands access, or specific neighborhood features are highly influential on local housing prices but were not available in the initial dataset.
 - **Future Work:** Enriching the dataset with economic data and geographically derived features (e.g., distance to Main Street, distance to Yellowstone National Park, neighborhood-level walkability/bikeability scores) could substantially boost predictive accuracy.

3.2.2. Model Robustness and Overfitting Risks

- **Single-Source Feature Encoding:** The strong correlation and influence of the *encoded_city_Big Sky* and *encoded_school_Lone Peak High School* variables, while accurate, suggest a potential over-reliance on a single geographic outlier. This can limit the model's generalizability to other, less exclusive areas of Gallatin County.
- **Interpretability vs. Performance:** While Lasso regression was chosen for its balance of performance and feature selection, the non-linear models (Polynomial Regression and SOM/Random Forest) indicated that complex, non-linear relationships exist. The current linear model structure may not fully capture these nuances.
 - **Future Work:** Future modeling could focus on advanced ensemble techniques and deep learning models (e.g., Neural Networks). These methods are better equipped to model the complex, non-linear interactions between dozens of features (structural, locational, and economic) while using techniques like early stopping and dropout to mitigate overfitting.

3.2.3. Generalizability and Predictive Scope

- **Geographic Scope:** The conclusions are strictly limited to single-family homes within Gallatin County, Montana. The findings (particularly the unique influence of Big Sky) are not transferable to other housing markets without significant modification.
 - **Future Work:** Extending the analysis to include other home types (condos, townhomes) or adjacent counties would broaden the scope, requiring careful re-evaluation of feature influence across different sub-markets.

In conclusion, our model provides a strong, interpretable foundation for understanding Gallatin County home prices, driven primarily by location (Big Sky) and property size. However, incorporating time-series analysis and enriching the feature set with economic and detailed amenity data represents the most critical steps for improving the model's predictive robustness and real-world applicability.